DATA STRUCTURES (DS) H/W PROGRAMS EXERCISE-1

1.Sum of two matrices

```
#include <stdio.h>

int main()
{
    int r, c;
    int a[10][10], b[10][10], sum[10][10];
    printf("Enter number of rows: ");
    scanf("%d", &r);

    printf("Enter number of columns: ");
    scanf("%d", &c);
    printf("\nEnter elements of Matrix A:\n");
    for (int i = 0; i < r; i++)
    {
        for (int j = 0; j < c; j++)
        {
            scanf("%d", &a[i][j]);
        }
    }
    printf("\nEnter elements of Matrix B:\n");
    for (int i = 0; i < r; i++)
    {
        for (int j = 0; j < c; j++)
        {
            scanf("%d", &b[i][j]);
        }
    }
    for (int i = 0; i < r; i++)
    {
```

```
    for (int j = 0; j < c; j++)
    {
        sum[i][j] = a[i][j] + b[i][j];
    }
}
printf("\nMatrix A:\n");
for (int i = 0; i < r; i++)
{
    for (int j = 0; j < c; j++)
    {
        printf("%d\t", a[i][j]);
    }
    printf("\n");
}
printf("\nMatrix B:\n");
for (int i = 0; i < r; i++)
{
    for (int j = 0; j < c; j++)
    {
        printf("%d\t", b[i][j]);
    }
    printf("\n");
}
printf("\nSum of Matrix A and Matrix B:\n");
for (int i = 0; i < r; i++)
{
    for (int j = 0; j < c; j++)
    {
        printf("%d\t", sum[i][j]);
    }
    printf("\n");
```

```
    }

    return 0;
}
```

OUTPUT

Enter number of rows: 2

Enter number of columns: 2

Enter elements of Matrix A:

1 2

3 4

Enter elements of Matrix B:

5 6

7 8

Matrix A:

1 2

3 4

Matrix B:

5 6

7 8

Sum of Matrix A and Matrix B:

6 8

10 12

2. Write a program in C to read n number of values in an array & display them in reverse order.

```
#include <stdio.h>

int main()
{
    int n, i;

    printf("Enter the number of elements: ");
    scanf("%d", &n);

    int arr[n];

    printf("Enter %d elements:\n", n);
    for(i = 0; i < n; i++)
```

```
    {
        scanf("%d", &arr[i]);
         }

    printf("\nArray in reverse order:\n");
    for(i = n - 1; i >= 0; i--)
    {
        printf("%d ", arr[i]);
    }

    printf("\n");

    return 0;
}
```

Formatted Output

```
Enter the number of elements: 5
Enter 5 elements:
10 20 30 40 50

Array in reverse order:
50 40 30 20 10
```

3. Count the total number of duplicate elements in an array.

```
#include <stdio.h>

int main()
{
    int n, i, j, count = 0;

    printf("Enter the number of elements: ");
    scanf("%d", &n);

    int arr[n];

    printf("Enter %d elements:\n", n);
    for(i = 0; i < n; i++)
    {
        scanf("%d", &arr[i]);
    }

    for(i = 0; i < n; i++)
    {
        for(j = i + 1; j < n; j++)
        {
            if(arr[i] == arr[j])
            {
                count++;
                break;
```

```
            }
        }
    }

    printf("\nTotal number of duplicate elements: %d\n", count);

    return 0;
}
```

Formatted Output
Enter the number of elements: 6
Enter 6 elements:
10 20 30 10 40 20

Total number of duplicate elements: 2

4. Print unique elements in an array.

```
#include <stdio.h>

int main()
{
    int n, i, j, count;

    printf("Enter the number of elements: ");
    scanf("%d", &n);

    int arr[n];

    printf("Enter %d elements:\n", n);
    for(i = 0; i < n; i++)
    {
        scanf("%d", &arr[i]);
    }

    printf("\nUnique elements:\n");

    for(i = 0; i < n; i++)
    {
        count = 0;

        for(j = 0; j < n; j++)
        {
            if(arr[i] == arr[j])
            {
                count++;
            }
        }

        if(count == 1)
        {
            printf("%d ", arr[i]);
        }
```

```
    }

    printf("\n");

    return 0;
}
```

Formatted Output

```
Enter the number of elements: 6
Enter 6 elements:
10 20 30 10 40 20

Unique elements:
30 40
```

5. Insert in sorted array.

```c
 #include <stdio.h>

int main()
{
    int n, i, value, pos;

    printf("Enter the number of elements: ");
    scanf("%d", &n);

    int arr[n + 1];

    printf("Enter %d elements in sorted order:\n", n);
    for(i = 0; i < n; i++)
    {
        scanf("%d", &arr[i]);
    }

    printf("Enter the element to insert: ");
    scanf("%d", &value);

    pos = 0;

    while(pos < n && arr[pos] < value)
    {
        pos++;
    }

    for(i = n; i > pos; i--)
    {
        arr[i] = arr[i - 1];
    }

    arr[pos] = value;
```

```
    printf("\nArray after insertion:\n");
    for(i = 0; i <= n; i++)
    {
        printf("%d ", arr[i]);
    }

    printf("\n");

    return 0;
}
```
Formatted Output

```
Enter the number of elements: 5
Enter 5 elements in sorted order:
10 20 30 40 50
Enter the element to insert: 35

Array after insertion:
10 20 30 35 40 50
```

6. Insert in unsorted array.

```
#include <stdio.h>

int main()
{
    int n, i, pos, value;

    printf("Enter the number of elements: ");
    scanf("%d", &n);

    int arr[n + 1];

    printf("Enter %d elements:\n", n);
    for(i = 0; i < n; i++)
    {
        scanf("%d", &arr[i]);
    }

    printf("Enter the element to insert: ");
    scanf("%d", &value);

    printf("Enter the position to insert: ");
    scanf("%d", &pos);

    for(i = n; i >= pos; i--)
    {
        arr[i] = arr[i - 1];
    }

    arr[pos - 1] = value;
```

```
    printf("\nArray after insertion:\n");
    for(i = 0; i <= n; i++)
    {
        printf("%d ", arr[i]);
    }

    printf("\n");

    return 0;
}
```

Formatted Output

```
Enter the number of elements: 5
Enter 5 elements:
40 10 50 20 30
Enter the element to insert: 25
Enter the position to insert: 4

Array after insertion:
40 10 50 25 20 30
```

7. Matrix multiplication.

```
#include <stdio.h>

int main()
{
    int r1, c1, r2, c2, i, j, k;

    printf("Enter rows and columns of first matrix: ");
    scanf("%d %d", &r1, &c1);

    printf("Enter rows and columns of second matrix: ");
    scanf("%d %d", &r2, &c2);

    if(c1 != r2)
    {
        printf("\nMatrix multiplication is not possible.\n");
        return 0;
    }

    int a[r1][c1], b[r2][c2], result[r1][c2];

    printf("\nEnter elements of first matrix:\n");
    for(i = 0; i < r1; i++)
    {
        for(j = 0; j < c1; j++)
        {
            scanf("%d", &a[i][j]);
        }
    }
```

```c
    printf("\nEnter elements of second matrix:\n");
    for(i = 0; i < r2; i++)
    {
        for(j = 0; j < c2; j++)
        {
            scanf("%d", &b[i][j]);
        }
    }

    for(i = 0; i < r1; i++)
    {
        for(j = 0; j < c2; j++)
        {
            result[i][j] = 0;

            for(k = 0; k < c1; k++)
            {
                result[i][j] += a[i][k] * b[k][j];
            }
        }
    }

    printf("\nResultant Matrix:\n");
    for(i = 0; i < r1; i++)
    {
        for(j = 0; j < c2; j++)
        {
            printf("%d ", result[i][j]);
        }
        printf("\n");
    }

    return 0;
}
```

Formatted Output

```
Enter rows and columns of first matrix: 2 2
Enter rows and columns of second matrix: 2 2

Enter elements of first matrix:
1 2
3 4

Enter elements of second matrix:
5 6
7 8

Resultant Matrix:
19 22
43 50
```

8. Matrix transpose.

```
#include <stdio.h>

int main()
{
   int r, c, i, j;

   printf("Enter the number of rows and columns: ");
   scanf("%d %d", &r, &c);

   int matrix[r][c];

   printf("\nEnter the elements of the matrix:\n");
   for(i = 0; i < r; i++)
   {
      for(j = 0; j < c; j++)
      {
         scanf("%d", &matrix[i][j]);
      }
   }

   printf("\nOriginal Matrix:\n");
   for(i = 0; i < r; i++)
   {
      for(j = 0; j < c; j++)
      {
         printf("%d ", matrix[i][j]);
      }
      printf("\n");
   }

   printf("\nTranspose of the Matrix:\n");
   for(i = 0; i < c; i++)
   {
      for(j = 0; j < r; j++)
      {
         printf("%d ", matrix[j][i]);
      }
      printf("\n");
   }

   return 0;
}
```

Formatted Output

```
Enter the number of rows and columns: 2 3

Enter the elements of the matrix:
1 2 3
4 5 6
```

Original Matrix:
1 2 3
4 5 6

Transpose of the Matrix:
1 4
2 5
3 6

9. Right diagonal sum.

```
#include <stdio.h>
int main()
{
    int n, i, j, sum = 0;

    printf("Enter the order of the matrix: ");
    scanf("%d", &n);

    int matrix[n][n];

    printf("\nEnter the elements of the matrix:\n");
    for(i = 0; i < n; i++)
    {
        for(j = 0; j < n; j++)
        {
            scanf("%d", &matrix[i][j]);
        }
    }

    for(i = 0; i < n; i++)
    {
        for(j = 0; j < n; j++)
        {
            if(i + j == n - 1)
            {
                sum += matrix[i][j];
            }
        }
    }

    printf("\nRight diagonal sum: %d\n", sum);

    return 0;
}
```

Formatted Output
Enter the order of the matrix: 3

Enter the elements of the matrix:
1 2 3

4 5 6
7 8 9

Right diagonal sum: 15

10. Check for identity matrix.

```c
#include <stdio.h>

int main()
{
   int n, i, j, flag = 1;

   printf("Enter the order of the matrix: ");
   scanf("%d", &n);

   int matrix[n][n];

   printf("\nEnter the elements of the matrix:\n");
   for(i = 0; i < n; i++)
   {
      for(j = 0; j < n; j++)
      {
         scanf("%d", &matrix[i][j]);
      }
   }

   for(i = 0; i < n; i++)
   {
      for(j = 0; j < n; j++)
      {
         if((i == j && matrix[i][j] != 1) ||
           (i != j && matrix[i][j] != 0))
         {
            flag = 0;
            break;
         }
      }

      if(flag == 0)
         break;
   }

   if(flag == 1)
      printf("\nThe matrix is an identity matrix.\n");
   else
      printf("\nThe matrix is not an identity matrix.\n");

   return 0;
}
```

Formatted output

CASE 1
Enter the order of the matrix: 3

Enter the elements of the matrix:
1 0 0
0 1 0
0 0 1

The matrix is an identity matrix.

CASE 2
Enter the order of the matrix: 3

Enter the elements of the matrix:
1 0 0
0 5 0
0 0 1

The matrix is not an identity matrix.

11. To find no. of vowels, consonants, digits and white spaces.

```
#include <stdio.h>

int main()
{
    char str[100];
    int i, vowels = 0, consonants = 0, digits = 0, spaces = 0;

    printf("Enter a string: ");
    fgets(str, sizeof(str), stdin);

    for(i = 0; str[i] != '\0'; i++)
    {
        if((str[i] >= 'a' && str[i] <= 'z') ||
           (str[i] >= 'A' && str[i] <= 'Z'))
        {
            if(str[i] == 'a' || str[i] == 'e' || str[i] == 'i' ||
               str[i] == 'o' || str[i] == 'u' ||
               str[i] == 'A' || str[i] == 'E' || str[i] == 'I' ||
               str[i] == 'O' || str[i] == 'U')
            {
                vowels++;
            }
            else
            {
                consonants++;
            }
        }
        else if(str[i] >= '0' && str[i] <= '9')
```

```
        {
            digits++;
        }
        else if(str[i] == ' ' || str[i] == '\t')
        {
            spaces++;
        }
    }

    printf("\nNumber of vowels: %d\n", vowels);
    printf("Number of consonants: %d\n", consonants);
    printf("Number of digits: %d\n", digits);
    printf("Number of white spaces: %d\n", spaces);

    return 0;
}
```

Formatted Output

```
Enter a string: Hello World 123

Number of vowels: 3
Number of consonants: 7
Number of digits: 3
Number of white spaces: 2
```

12. Frequency of a character in a string.

```
#include <stdio.h>
int main()
{
    char str[100], ch;
    int i, count = 0;

    printf("Enter a string: ");
    fgets(str, sizeof(str), stdin);

    printf("Enter the character to find frequency: ");
    scanf("%c", &ch);

    for(i = 0; str[i] != '\0'; i++)
    {
        if(str[i] == ch)
        {
            count++;
        }
    }

    printf("\nFrequency of '%c': %d\n", ch, count);

    return 0;
}
```

Formatted Output

Enter a string: programming
Enter the character to find frequency: m

Frequency of 'm': 2

13. Count words in a string.

```c
#include <stdio.h>

int main()
{
   char str[100];
   int i, words = 0;

   printf("Enter a string: ");
   fgets(str, sizeof(str), stdin);

   for(i = 0; str[i] != '\0'; i++)
   {
      if(str[i] != ' ' && str[i] != '\n' &&
        (i == 0 || str[i - 1] == ' '))
      {
         words++;
      }
   }

   printf("\nNumber of words: %d\n", words);

   return 0;
}
```

Formatted Output

Enter a string: C programming is easy

Number of words: 4

14. Sort string array .

sorted an array of strings using simple bubble sort and strcmp() function.

```c
#include <stdio.h>
#include <string.h>

int main()
{
   int n, i, j;
   char temp[100];

   printf("Enter the number of strings: ");
   scanf("%d", &n);
```

```
    char str[n][100];

    printf("Enter %d strings:\n", n);
    for(i = 0; i < n; i++)
    {
        scanf("%s", str[i]);
    }

    for(i = 0; i < n - 1; i++)
    {
        for(j = 0; j < n - i - 1; j++)
        {
            if(strcmp(str[j], str[j + 1]) > 0)
            {
                strcpy(temp, str[j]);
                strcpy(str[j], str[j + 1]);
                strcpy(str[j + 1], temp);
            }
        }
    }

    printf("\nStrings in sorted order:\n");
    for(i = 0; i < n; i++)
    {
        printf("%s\n", str[i]);
    }

    return 0;
}
```

Formatted Output

```
Enter the number of strings: 5
Enter 5 strings:
Mango
Apple
Orange
Banana
Grapes

Strings in sorted order:
Apple
Banana
Grapes
Mango
Orange
```

15. Toggle case of a sentence. (lowercase to uppercase & vice versa)

```
#include <stdio.h>
```

```
int main()
{
   char str[100];
   int i;

   printf("Enter a sentence: ");
   fgets(str, sizeof(str), stdin);

   for(i = 0; str[i] != '\0'; i++)
   {
      if(str[i] >= 'a' && str[i] <= 'z')
      {
         str[i] = str[i] - 32;
      }
      else if(str[i] >= 'A' && str[i] <= 'Z')
      {
         str[i] = str[i] + 32;
      }
   }

   printf("\nToggled sentence:\n");
   printf("%s", str);

   return 0;
}
```

Formatted Output

Enter a sentence: Hello World

Toggled sentence:
hELLO wORLD

For example:
Input → Hello WORLD 123
Output → hELLO world 123